1. Difference between Basic Filter, Advanced Filter, and Slicer

Filter Type	Description
Basic Filter	Allows selecting specific values from a list to filter data (e.g., select only 2023).
Advanced Filter	Allows applying conditions like *greater than, less than, contains,* etc.
Slicer	A visual filter placed on the report canvas that allows users to interactively filter visuals.

Example:
Basic Filter → Select Year = 2023
Advanced Filter → Sales > 50000
Slicer → Region selection on report page.

---

2. Apply a Basic Filter to display only sales for 2023

Steps in Power BI

1. Open **Power BI Desktop**
2. Load the **Sales Dataset**
3. Go to **Filters Pane**
4. Drag **Date/Year column** into **Filters on this page**
5. Select **Basic Filtering**
6. Tick **2023**

Result → All visuals show **Sales data only for 2023**.

---

3. Create Pie Chart of Sales by Product with Region Slicer

Steps

1. Select **Pie Chart Visual**
2. Drag:
    - **Product → Legend**
    - **Sales → Values**

3. Insert a **Slicer Visual**
4. Drag **Region → Slicer field**

Result → Users can select a **Region**, and the **Pie Chart updates automatically**.

---

4. Visual-Level vs Page-Level vs Report-Level Filters

Filter Type	Applies To
Visual-Level Filter	Only one specific visual
Page-Level Filter	All visuals on a single page
Report-Level Filter	All pages and visuals in the report

Example:
Visual Filter → Sales > 10000 on one chart only.

---

5. Apply Report-Level Filter for India Sales

Steps

1. Go to **Filters Pane**
2. Drag **Country column** into **Filters on all pages**
3. Select **India**

Result → Entire report shows **Sales data only for India**.

---

6. Role of Drill-Down in Power BI

Drill-Down allows users to explore data from **higher level to detailed level**.

Example:

Sales → Year → Quarter → Month → Day

Benefits:

- Detailed data analysis
- Better data exploration
- Identify trends easily

---

7. Create Hierarchy Drill-Down (Country → Region → Product Category)

Steps

1. In **Fields Pane**
2. Right-click **Country → Create Hierarchy**
3. Add:
    - Region
    - Product Category
4. Use this hierarchy in a **Bar Chart**

Result → Users can **drill from Country → Region → Product Category**.

---

8. Implement Drill-Through for Product Details

Steps

1. Create a **New Page (Product Details)**
2. Drag **Product** into **Drill-through Filters**
3. Add visuals like:
    - Region
    - Sales
4. On main page:
    - Right click **Product**
    - Select **Drill Through → Product Details**

Result → Shows **region-wise sales for selected product**.

---

9. Add Slicer for Product Category

Steps

1. Insert **Slicer Visual**
2. Drag **Product Category**
3. Select any category

Result →
All visuals + drill-down hierarchy **update based on selected category**.

---

10. Difference Between Drill-Down and Drill-Through

Feature	Drill-Down	Drill-Through
Purpose	Explore levels of hierarchy	Navigate to another report page
Example	Year → Quarter → Month	Product → Product Details Page

Real-world Example:

Drill-Down → Sales by Year → Month
Drill-Through → Click Product → See Product performance page.

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11. Analyze Sales Trends (Year → Quarter → Month → Day)

Steps

1. Insert **Line Chart**
2. Drag:
    - **Date Hierarchy → Axis**
    - **Sales → Values**
3. Enable **Drill Down**

Result → Users can view **sales trend from Year → Day level**.

---

12. Measure-Based Filter (Sales > ₹50,000)

Step 1: Create Measure

Total Sales = SUM(Sales[Sales])

Step 2: Apply Filter

1. Drag **Total Sales measure** into **Filters Pane**
2. Set condition:

Total Sales > 50000

Result → Only products with **sales greater than ₹50,000 appear**.

---

13. Keep All Filters in Drill-Through

Keep All Filters ensures that **existing filters and slicer selections are passed to the drill-through page**.

Example:

If filters applied:

- Country = India
- Region = North

After drill-through → The **same filters remain active** on the detail page.

---

14. Create Custom Drill-Through Page for Region Sales

Steps

1. Create **New Page → Region Details**
2. Drag **Region → Drill-through field**
3. Add visuals:
    - Line Chart (Sales vs Date)
    - Table (Product, Sales)

Result → Selecting a region shows **sales trends over time**.

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15. Design Interactive Power BI Dashboard

Dashboard Components:

1️⃣ **Slicers**

- Region
- Product Category
- Year

2️⃣ **Visuals**

- Pie Chart → Sales by Product
- Bar Chart → Sales by Region

- Line Chart → Sales Trend

3️⃣ **Features**

- Drill-Down hierarchy (Country → Region → Category)
- Drill-Through pages (Product Details)
- Report Level Filter (Country = India)

Result → A **fully interactive sales analysis dashboard**.